

Nicholas Swisher

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SUMMARY

New graduate software engineer with experience building and deploying API-driven applications. Built a full-stack recommendation system using React, TypeScript, and FastAPI, with a focus on clear API design, data modeling, and core computer science fundamentals.

EDUCATION

University of Colorado Boulder

Boulder, CO

B.S. Applied Computer Science — GPA: 3.74 — Cum Laude

Dec 2025

University of North Texas

Denton, TX

B.S. Kinesiology — GPA: 3.6 — Cum Laude

May 2020

PROJECTS

Content Recommendation System

React, TypeScript, Python, FastAPI, SQLite

- Developed and deployed a full-stack recommendation application on Render using a React/TypeScript frontend and FastAPI backend.
- Implemented a content-based recommendation engine using TF-IDF and cosine similarity over a dataset of 45,000+ content items.
- Designed a structured SQLite data layer and REST API to serve metadata and recommendation results.

Event Management Platform

C++

- Developed a modular backend system modeling users, events, and transaction workflows using object-oriented design principles.
- Applied Factory and Singleton patterns to support extensibility and centralized application coordination.

Data Structures and Benchmarking

C++

- Implemented templated HashMap, BST, and Linear List structures with full Catch2 unit test coverage.
- Benchmarked runtime performance across datasets to evaluate algorithmic tradeoffs.
- Automated unit test execution with GitHub Actions CI to validate correctness across changes.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, SQL, JavaScript

Frameworks & Libraries: React, FastAPI, scikit-learn, pandas

Databases: SQLite, MongoDB

Concepts: OOP, Data Structures & Algorithms, REST APIs, Test-Driven Development (TDD), Unit Testing (Catch2)

Tools: Git, GitHub Actions, VS Code, Render

EMPLOYMENT

Safeway

Front End Manager

2024 – 2025

- Led front-end operations, resolving customer issues and coordinating team workflows.

Spectrum

Account Manager

2023 – 2024

- Managed client communication and issue resolution while supporting account operations.

Rocky Mountain Cycle Plaza

Technician

2021 – 2023

- Diagnosed and repaired complex mechanical systems while communicating findings to customers.